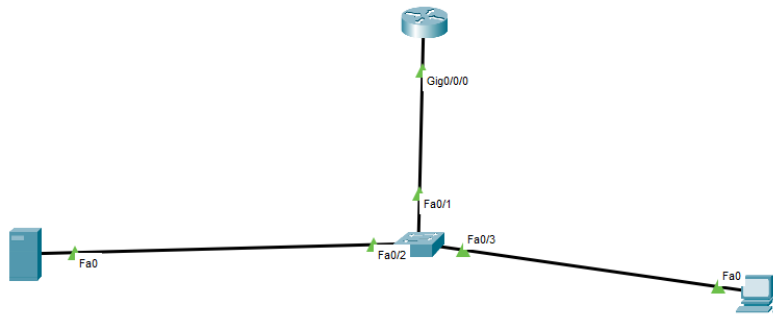


Complete AAA Lab Configuration: TACACS+, RADIUS, and SSH Fallback

1. Network Topology & IP Address Plan

Connect all devices to a central Layer 2 Switch (e.g., a Cisco 2960).

- **Router (R1)** connected to Switch Port Gi0/0/0
- **TACACS+ Server** connected to Switch Port F0/2
- **PC (Admin)** connected to Switch Port F0/3



Device	Interface	IP Address	Subnet Mask	Default Gateway
Router (R1)	G0/0	192.168.10.1	255.255.255.0	N/A
Switch (SW1)	VLAN 1	192.168.10.2	255.255.255.0	192.168.10.1
TACACS+ Server	NIC	192.168.10.10	255.255.255.0	192.168.10.1
RADIUS Server	NIC	192.168.10.20	255.255.255.0	192.168.10.1
PC (Admin)	NIC	192.168.10.100	255.255.255.0	192.168.10.1

2. PC and Server Configuration (Packet Tracer GUI)

PC (Admin) Setup

1. Open the PC -> **Desktop** tab -> **IP Configuration**.
2. Set IPv4 Address: 192.168.10.100
3. Subnet Mask: 255.255.255.0
4. Default Gateway: 192.168.10.1

TACACS+ Server Setup

1. Open Server -> **Desktop** -> **IP Configuration**.
2. IP: 192.168.10.10, Mask: 255.255.255.0, Gateway: 192.168.10.1
3. Go to **Services** tab -> **AAA**. Turn Service **ON**.
4. **Network Setup**: Client Name: R1, Client IP: 192.168.10.1, Secret: TacacsKey123, Server Type: TACACS. Click **Add**.
5. **User Setup**: Username: bhaskar, Password: SecurePass1!. Click **Add**.

RADIUS Server Setup

1. Open Server -> **Desktop** -> **IP Configuration**.
2. IP: 192.168.10.20, Mask: 255.255.255.0, Gateway: 192.168.10.1
3. Go to **Services** tab -> **AAA**. Turn Service **ON**.
4. **Network Setup**: Client Name: R1, Client IP: 192.168.10.1, Secret: RadiusKey123, Server Type: RADIUS. Click **Add**.
5. **User Setup**: Username: bhaskar, Password: SecurePass1!. Click **Add**.

R1 config

!

hostname R1

!

!

!

enable secret 5 \$1\$mERr\$hx5rVt7rPNoS4wqbXKX7m0

!

!

!

!

!

aaa new-model

!

aaa authentication login SSH-AUTH group tacacs+ group radius local

!

!

aaa authorization exec SSH-AUTH group tacacs+ group radius local

aaa authorization exec default group tacacs+ group radius local

ip cef

no ipv6 cef

!

!

!

username admin privilege 15 secret 5 \$1\$mERr\$vTbHul1N28cEp8lkLqr0f/

!

!

ip ssh version 2

ip domain-name enterprise.local

```
!  
!  
spanning-tree mode pvst  
!  
!  
interface GigabitEthernet0/0/0  
ip address 192.168.10.1 255.255.255.0  
duplex auto  
speed auto  
  
tacacs-server host 192.168.10.10 key TacacsKey123  
tacacs-server key TacacsKey123  
radius server RADIUS-SRV  
address ipv4 192.168.10.10 auth-port 1645  
key RadiusKey123  
radius server 192.168.10.10  
address ipv4 192.168.10.10 auth-port 1645  
key RadiusKey123  
line vty 0 4  
login authentication SSH-AUTH  
transport input ssh  
end
```

3. Switch (SW1) Configuration

```
configure terminal
```

! Configure Management IP

interface vlan 1

ip address 192.168.10.2 255.255.255.0

no shutdown

exit

ip default-gateway 192.168.10.1

interface range f0/1 - 4

switchport mode access

spanning-tree portfast

exit

SW1> enable

SW1# configure terminal

! Enable AAA and define the RADIUS server

```
SW1(config)# aaa new-model
```

```
SW1(config)# radius-server host 192.168.10.20 key RadiusKey123
```

! Define the 802.1X authentication method to use RADIUS

```
SW1(config)# aaa authentication dot1x default group radius
```

! Enable 802.1X globally on the switch

```
SW1(config)# dot1x system-auth-control
```

! Apply 802.1X to the port connected to the PC (F0/3)

```
SW1(config)# interface FastEthernet0/3
```

```
SW1(config-if)# switchport mode access
```

```
SW1(config-if)# dot1x port-control auto
```

```
SW1(config-if)# end
```